

United States Federal Automotive Vibration Compliance Standards

This document defines the mandatory vibration compliance thresholds for automotive component validation under simulated environmental exposure conditions. All vendor-submitted test reports must comply with the limits specified below. Non-compliance may result in batch rejection and financial penalties.

Section 1: Water Exposure Test

Maximum allowable vibration level: 4.5 mm/s

Any individual component exceeding this limit shall be classified as non-compliant.

Section 2: Sand/Dust Exposure Test

Maximum allowable vibration level: 5.0 mm/s

Components exceeding the specified limit shall require corrective validation.

Section 3: Mud Exposure Test

Maximum allowable vibration level: 4.0 mm/s

Any deviation beyond the permissible limit shall be deemed a compliance failure.

Section 4: Batch-Level Rejection Criteria

If more than 20 percent of tested components exceed the applicable vibration threshold for a given environmental condition, the entire batch shall be rejected.